

EXPERIMENT 21: PERFORM SHARPENING OF IMAGE USING LAPLACIAN MASK IMPLEMENTED WITH AN EXTENSION OF DIAGONAL NEIGHBORS

Aim

To sharpen an image using a Laplacian mask with diagonal neighbors (8-neighborhood) in Python and OpenCV.

Algorithm

1. Import OpenCV and NumPy libraries.
2. Read the input image.
3. Convert the image to grayscale.
4. Define the 8-neighbor Laplacian kernel.
5. Apply the Laplacian filter using filter2D().
6. Subtract the Laplacian image from the original image.
7. Save and display the sharpened image.

Laplacian Mask

```
1  1  1
1 -8  1
1  1  1
```

Python Code

```
import cv2
import numpy as np

img = cv2.imread("input.jpg")

if img is None:
    print("Image not found")
    exit()

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

```
kernel = np.array([[1, 1, 1],  
                  [1,-8, 1],  
                  [1, 1, 1]], dtype=np.float32)  
  
laplacian = cv2.filter2D(gray, cv2.CV_32F, kernel)  
  
sharpened = np.clip(gray - laplacian, 0, 255).astype(np.uint8)  
  
cv2.imwrite("laplacian_8neighbor_output.jpg", sharpened)  
  
cv2.imshow("Input Image", gray)  
cv2.imshow("Sharpened Image", sharpened)  
  
cv2.waitKey(0)  
cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the image was successfully sharpened using the 8-neighbor Laplacian mask with diagonal neighbors.